

Python Programming Fundamentals

Strings and Slicing

Programming Fundamentals Team

SETU

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Outline

1. Strings Are Sequences
2. Slicing Lists
3. Slicing Strings
4. String Methods
5. Splitting and Joining



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1. Strings Are Sequences

A string is a sequence of characters.

```
course = "Python"

print(course[0])
print(course[1])
print(course[-1])
```



1. Strings Are Sequences

A string is a sequence of characters.

```
course = "Python"

print(course[0])
print(course[1])
print(course[-1])
```

Output:

```
P
y
n
```



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2. Slicing Lists

Slicing extracts part of a list.

```
numbers = [10, 20, 30, 40, 50, 60]

print(numbers[0:3]) #[10, 20, 30]
print(numbers[:3]) #[10, 20, 30]
print(numbers[3:]) #[40, 50, 60]
print(numbers[-2:])#[50, 60]
```



2. Slicing Lists

Slicing extracts part of a list.

```
numbers = [10, 20, 30, 40, 50, 60]

print(numbers[0:3]) #[10, 20, 30]
print(numbers[:3]) #[10, 20, 30]
print(numbers[3:]) #[40, 50, 60]
print(numbers[-2:])#[50, 60]
```

Remember: the stop position is not included when using :n

However , the start position is included when using n:

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3. Slicing Strings

The same idea works with strings.

```
word = "Programming"

print(word[:4]) #Prog
print(word[4:]) #ramming
print(word[-3:]) #ing
print(word[::-1]) #gnimmargorP
```



3. Slicing Strings

The same idea works with strings.

```
word = "Programming"

print(word[:4]) #Prog
print(word[4:]) #ramming
print(word[-3:]) #ing
print(word[::-1]) #gnimmargorP
```

Useful for:

- extracting initials
- checking prefixes
- reversing text
- splitting codes or identifiers



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4. String Methods

```
name = "  siobhan roche  "

print(name.strip()) #siobhan roche
print(name.upper()) #  SIOBHAN ROCHE
print(name.title()) #  Siobhan Roche
print(name.replace("roche", "Roche")) #  siobhan Roche
```



4. String Methods

```
name = "  siobhan roche  "

print(name.strip()) #siobhan roche
print(name.upper()) #  SIOBHAN ROCHE
print(name.title()) #  Siobhan Roche
print(name.replace("roche", "Roche")) #  siobhan Roche
```

String methods return a new string.

They do not change the original string in place.



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5. Splitting and Joining

```
sentence = "Python is fun"
words = sentence.split()
print(words)
```



5. Splitting and Joining

```
sentence = "Python is fun"
words = sentence.split()
print(words)
```

```
words = ["Python", "is", "fun"]
sentence = " ".join(words)
print(sentence)
```



Thanks for Watching - Any questions?

